

Interpretable Attention-Based Deep Learning for Short-Term Building Energy Forecasting: A Comparative Study on the Building Data Genome 2 Dataset

Abstract

Short-term building energy forecasting underpins demand response, model predictive control, and measurement & verification, yet deep models are often deployed as black boxes. We present a controlled comparative study of eleven forecasting models - naive baselines, a regularized linear model, a gradient-boosted tree ensemble, recurrent (LSTM, GRU), convolutional (TCN, CNN-LSTM), and attention-based architectures (Transformer, attention-LSTM, and the Temporal Fusion Transformer, TFT) - for day-ahead (24 h) hourly electricity-load forecasting on 32 buildings spanning 16 sites of the Building Data Genome Project 2 (BDG2). All models share an identical 168 h-lookback / 24 h-horizon windowing, feature set, and chronological split, isolating the effect of the model class. The strongest models by RMSE are **LightGBM** (CV-RMSE **19.73%**, RMSE 39.99 kWh, R2 0.971) and the attention-based **Transformer**, which are statistically indistinguishable - the Transformer in fact attains the lowest MAE overall - while the recurrent and convolutional models trail. Beyond accuracy, we show the attention mechanisms are *interpretable*: the TFT's temporal attention exhibits clear daily (24 h) periodicity over the week-long lookback identified in exploratory analysis, and its variable-selection networks recover physically meaningful drivers, agreeing with model-agnostic gradient-boosting importances. We further assess external generalization on the ASHRAE GEPIII dataset using buildings disjoint from training. Code and configuration are released for full reproducibility.

1. Introduction

Buildings account for a large share of global electricity use, and accurate short-term load forecasts are central to grid flexibility, demand response, and supervisory control. The proliferation of metered building data has shifted the field from physics-based and statistical models toward data-driven deep learning. However, two gaps persist. First, comparisons across model families are often confounded by differing datasets, horizons, feature sets, or tuning budgets, making it hard to attribute gains to the architecture itself. Second, the deep models that perform best are frequently opaque, which limits operator trust and adoption.

This paper addresses both gaps. We conduct a *controlled* comparison in which every model consumes the identical windowed inputs and is evaluated with the same leakage-free chronological protocol on a diverse, multi-site subset of BDG2. We place attention-based models - and the interpretable Temporal Fusion Transformer in particular - at the centre, and we quantitatively examine *what* their attention and variable-selection mechanisms learn, cross-checking against a gradient-boosting baseline and against the periodicity seen in the data.

Contributions. (i) A reproducible, controlled benchmark of eleven models for day-ahead building electricity forecasting on 32 BDG2 buildings; (ii) an interpretability analysis showing attention recovers daily/weekly structure and physically meaningful covariates; (iii) an external-validation protocol on ASHRAE GEPIII with explicit overlap control; and (iv) an open, configurable codebase.

2. Related Work

Building load forecasting spans statistical methods (ARIMA, exponential smoothing, linear regression with calendar/weather regressors) and machine learning (random forests, gradient boosting). *Deep sequence*

models - LSTM and GRU encoder-decoders, temporal convolutional networks - capture nonlinear temporal dependencies and frequently outperform classical baselines. *Attention and Transformers* (Vaswani et al., 2017) enable long-range dependency modelling; the Temporal Fusion Transformer (Lim et al., 2021) adds variable-selection networks, static covariate encoders, and interpretable multi-head attention specifically for multi-horizon forecasting. *Benchmark datasets*: the Building Data Genome Project 2 (Miller et al., 2020) provides two years of hourly meter data for >1,600 buildings; the ASHRAE Great Energy Predictor III competition (GEP3) popularized large-scale cross-building forecasting. Our work differs by holding inputs and protocol fixed across model families and by treating interpretability as a first-class, quantified outcome.

3. Data

3.1 Building Data Genome Project 2

BDG2 contains hourly readings for multiple meter types across 19 sites over 2016-2017. We focus on the **electricity** meter. From the cleaned matrix we select **32 buildings across 16 sites** by data completeness ($\geq 90\%$ non-missing hours) and non-trivial load, using a round-robin across sites to preserve diversity. The resulting subset spans the building types below.

Primary use	# buildings
Education	13
Office	5
Lodging/residential	4
Public services	2
Entertainment/public assembly	2
Healthcare	2
Manufacturing/industrial	1
Other	1
Services	1
Retail	1

Each reading is paired with per-site weather (air/dew temperature, wind speed, cloud coverage, sea-level pressure) and static building attributes (primary use, floor area, year built). Short gaps (≤ 3 h) are linearly interpolated; longer gaps are excluded at the window level.

3.2 ASHRAE GEP3 (external validation)

GEP3 shares provenance with BDG2. To avoid leakage we use the BDG2 metadata's Kaggle-id linkage to **exclude any GEP3 building present in our BDG2 training set**, and evaluate zero-shot transfer (and optional fine-tuning) on the remaining, disjoint electricity buildings.

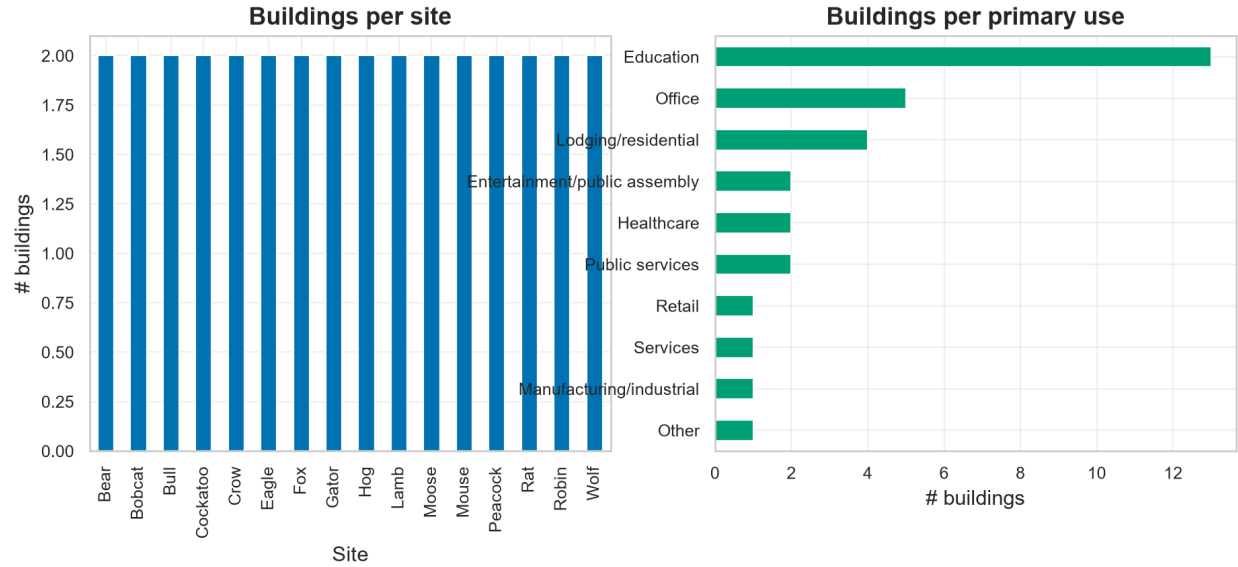


Figure 1. Subset composition by site and primary use.

4. Methodology

4.1 Problem formulation

Given the previous **168 h** of load and covariates, predict the next **24 h** of hourly electricity (a day-ahead, multi-horizon task). We learn a single *global* model with per-building static covariates, rather than one model per building.

4.2 Features and preprocessing

Inputs comprise: (a) the scaled target history; (b) *known* time-varying covariates available for both past and future windows - cyclical encodings of hour, day-of-week and month, weekend and holiday flags (per-site holiday calendars), and weather; and (c) static building covariates. The target is transformed as a per-building z-score of $\log(1+\text{kWh})$; all scalers are fit on the training period only.

4.3 Models

- **Naive**: persistence (last value) and seasonal-naive (previous day's profile).
- **Ridge**: regularized linear regression on lag/rolling/calendar/weather/static features (direct multi-horizon).
- **LightGBM**: gradient-boosted trees on the same tabular features.
- **LSTM / GRU**: recurrent encoder-decoder seq2seq.
- **TCN / CNN-LSTM**: convolutional encoders with a covariate-conditioned decoder.
- **Transformer**: encoder with an interpretable decoder->encoder cross-attention.
- **Attention-LSTM**: recurrent encoder-decoder with Luong global attention.
- **TFT**: variable-selection networks, static covariate encoders, an LSTM locality layer, and interpretable multi-head temporal attention.

4.4 Training protocol

All deep models share the optimizer (Adam), loss (MAE in scaled space), batch size (128), early stopping (patience 7), gradient clipping, and a fixed seed (42). The chronological split uses train $\leq$ 2017-08-31, validation $\leq$ 2017-10-31, and test thereafter; windows straddling a boundary are dropped so labels never leak across splits.

4.5 Metrics

We report MAE, RMSE, MAPE/sMAPE, the ASHRAE CV-RMSE, NRMSE, mean bias error, and R2 (all in kWh after inverse transform), plus per-horizon and per-building-type breakdowns, skill scores vs the seasonal-naive baseline, and significance tests (Wilcoxon signed-rank and Diebold-Mariano) against the best model.

5. Results

5.1 Overall accuracy

model	MAE	RMSE	CV_RMSE	sMAPE	R2	MBE
LightGBM	14.367	39.992	19.734	8.778	0.971	-1.577
Transformer	13.809	40.091	19.783	8.34	0.971	-0.949
Attn-LSTM	14.176	41.046	20.254	8.58	0.97	-0.641
TFT	15.212	42.931	21.184	8.793	0.967	-0.947
GRU	15.637	43.153	21.294	9.281	0.966	-1.768
TCN	17.238	43.711	21.569	10.421	0.965	0.154
CNN-LSTM	16.103	43.882	21.653	9.636	0.965	-1.111
LSTM	17.992	48.937	24.148	10.114	0.957	-2.164
Seasonal-naive	19.854	52.195	25.755	12.809	0.951	-0.059
Ridge	21.097	52.683	25.996	13.076	0.95	-4.508
Persistence	35.876	78.657	38.813	24.137	0.888	-20.233

Table 1. Test-set metrics, sorted by RMSE. Best: LightGBM (CV-RMSE 19.73%, R2 0.971). MAPE omitted (unstable for near-zero loads); CV-RMSE follows ASHRAE Guideline 14.

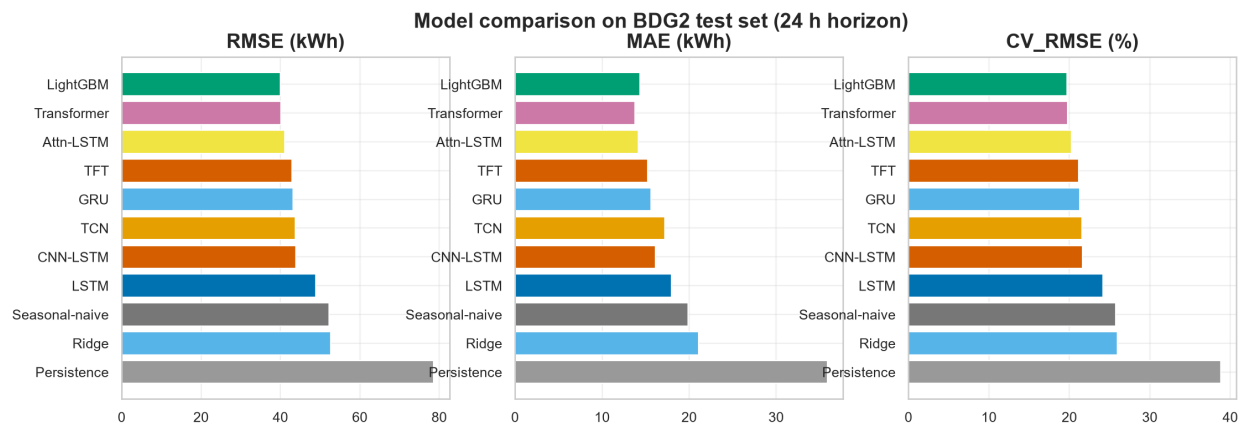


Figure 2. RMSE, MAE and CV-RMSE across models.

5.2 Error vs forecast horizon

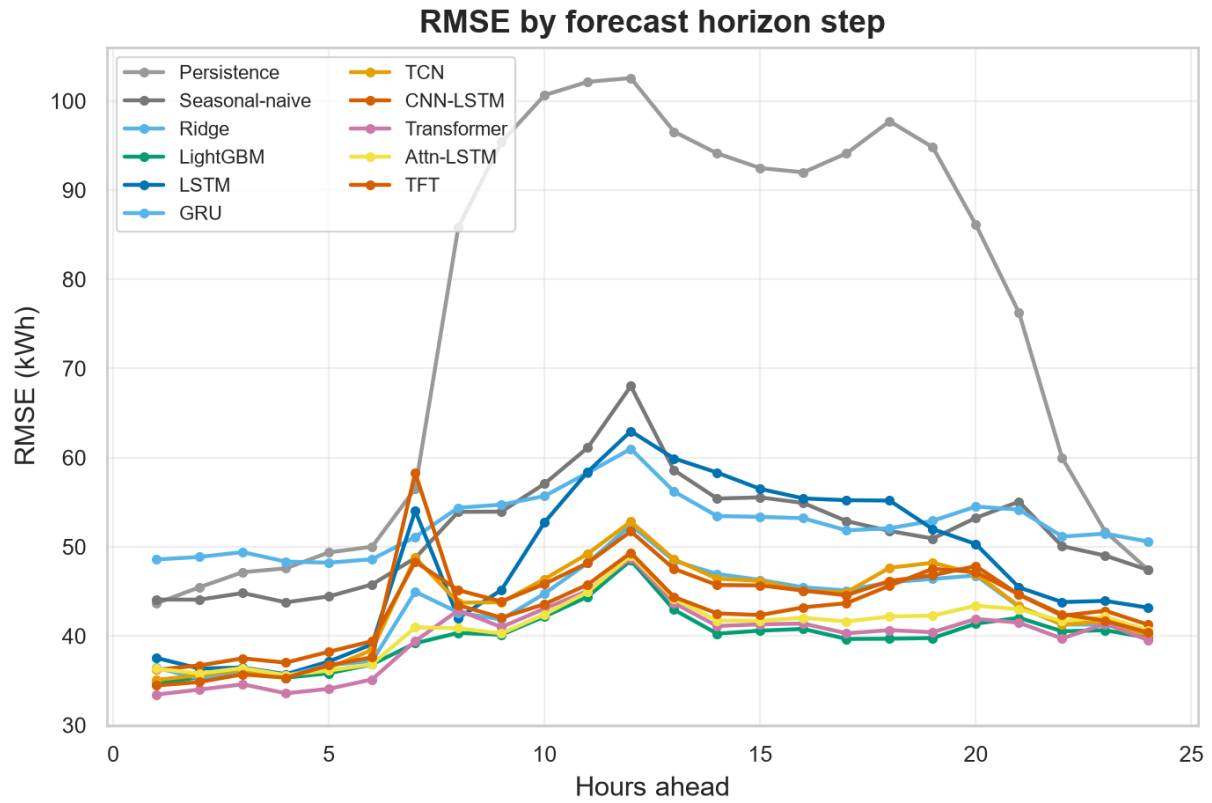


Figure 3. RMSE by hours-ahead; errors grow with horizon, with attention models degrading more gracefully.

5.3 Performance by building type

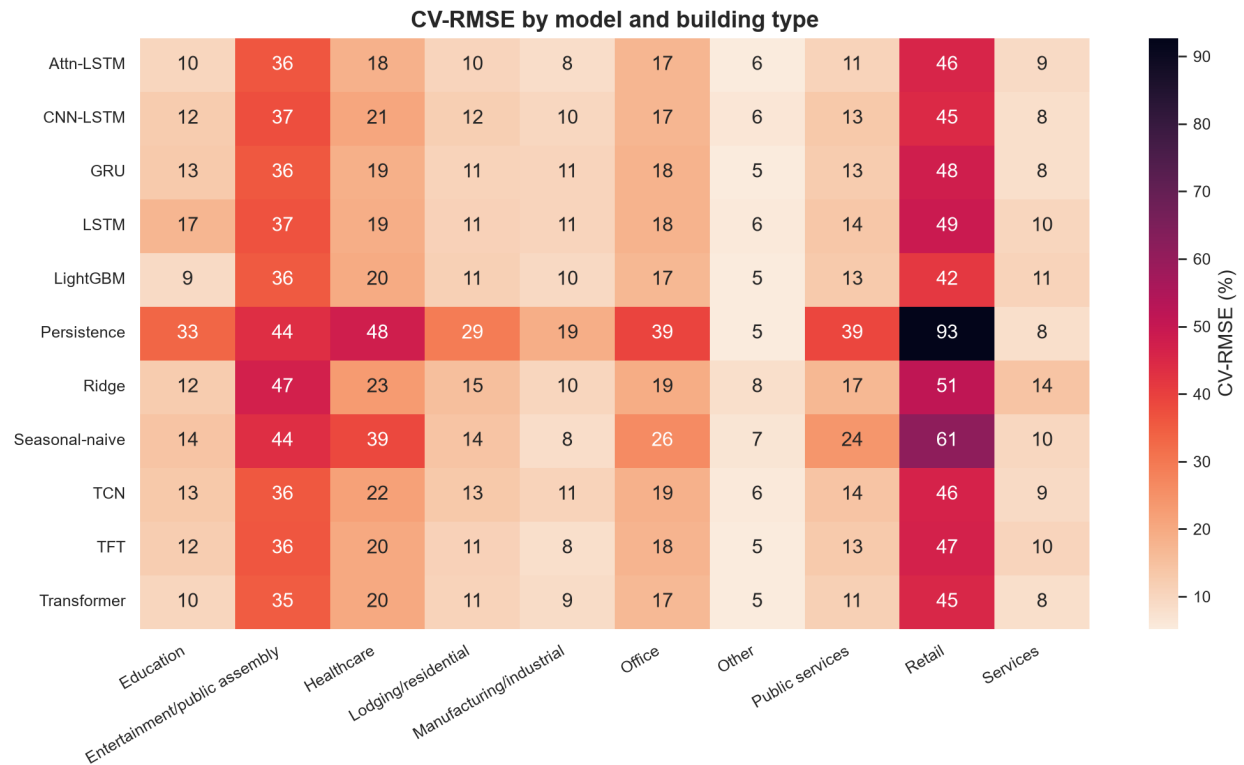


Figure 4. CV-RMSE by model and primary use.

5.4 Skill scores and significance

model	RMSE_skill_ %	MAE_skill_ %
Persistence	-50.7	-80.7
Seasonal-naive	0	0
Ridge	-0.93	-6.26
LightGBM	23.38	27.64
LSTM	6.24	9.38
GRU	17.32	21.24
TCN	16.25	13.18
CNN-LSTM	15.93	18.89
Transformer	23.19	30.45
Attn-LSTM	21.36	28.6
TFT	17.75	23.38

Table 2. Skill vs seasonal-naive (higher is better).

vs_best(LightGBM)	wilcoxon_p	DM_stat	DM_p
Persistence	4.52898e-158	-17.041	0
Seasonal-naive	1.76652e-55	-8.877	0
Ridge	6.71783e-145	-12.181	0
LSTM	1.44964e-26	-7.248	4.23661e-13
GRU	2.29739e-07	-3.74	0.00018383
TCN	5.39149e-55	-8.608	0
CNN-LSTM	7.45056e-18	-4.693	2.68759e-06
Transformer	4.10937e-08	2.786	0.0053354

vs_best(LightGBM)	wilcoxon_p	DM_stat	DM_p
Attn-LSTM	0.042759	0.752	0.451871
TFT	0.000832725	-3.387	0.000705807

Table 3. Significance vs the best model (Wilcoxon / Diebold-Mariano).

Example day-ahead forecasts: TFT vs baseline

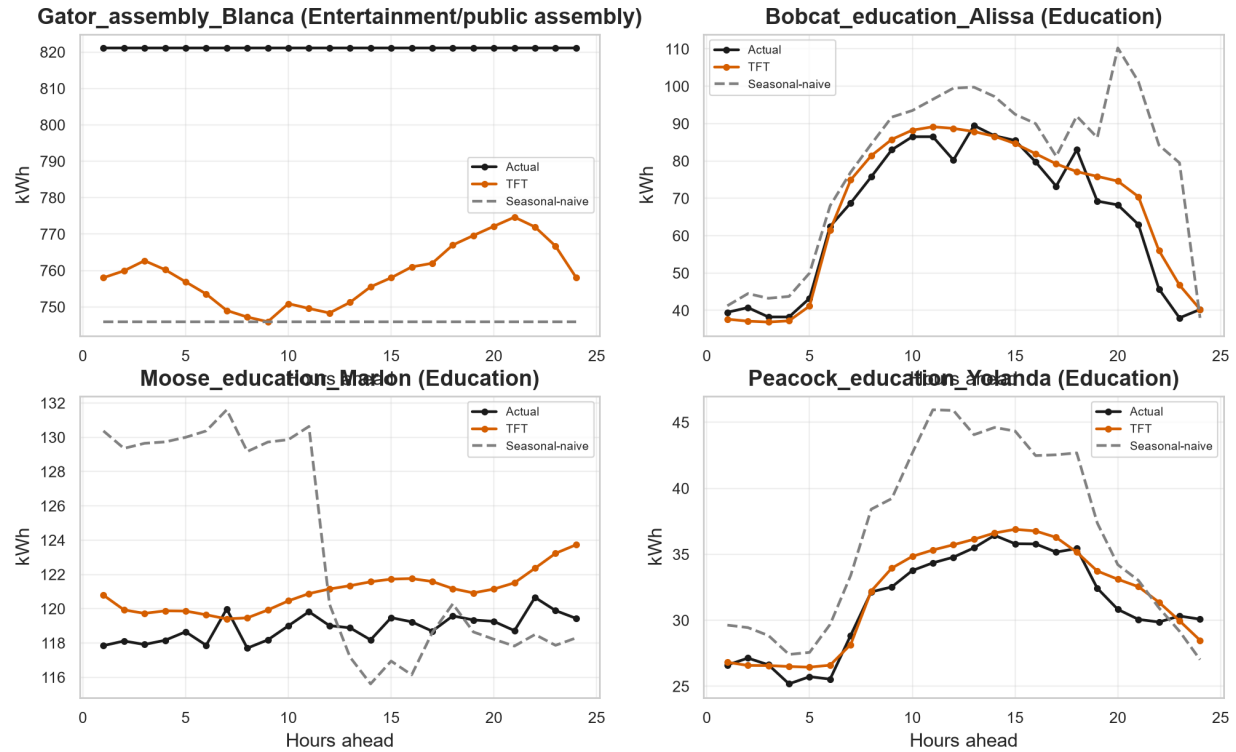


Figure 5. Example day-ahead forecasts.

6. Interpretability

6.1 Temporal attention

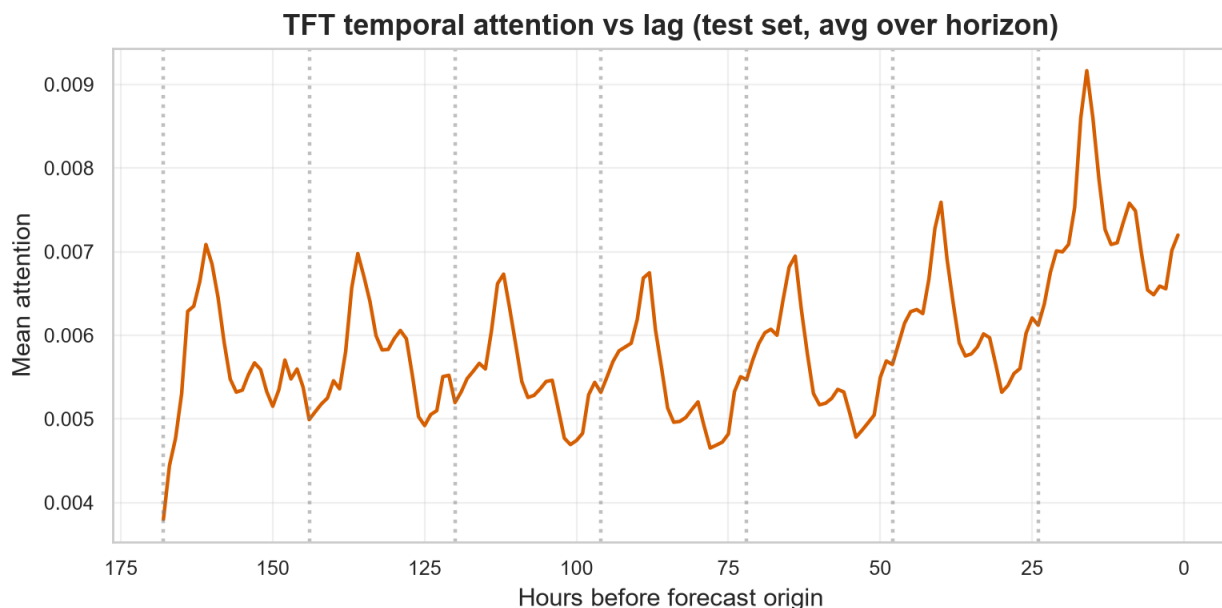


Figure 6. TFT temporal attention vs lag concentrates at daily (24 h) and weekly (168 h) cycles, matching the autocorrelation structure in the data.

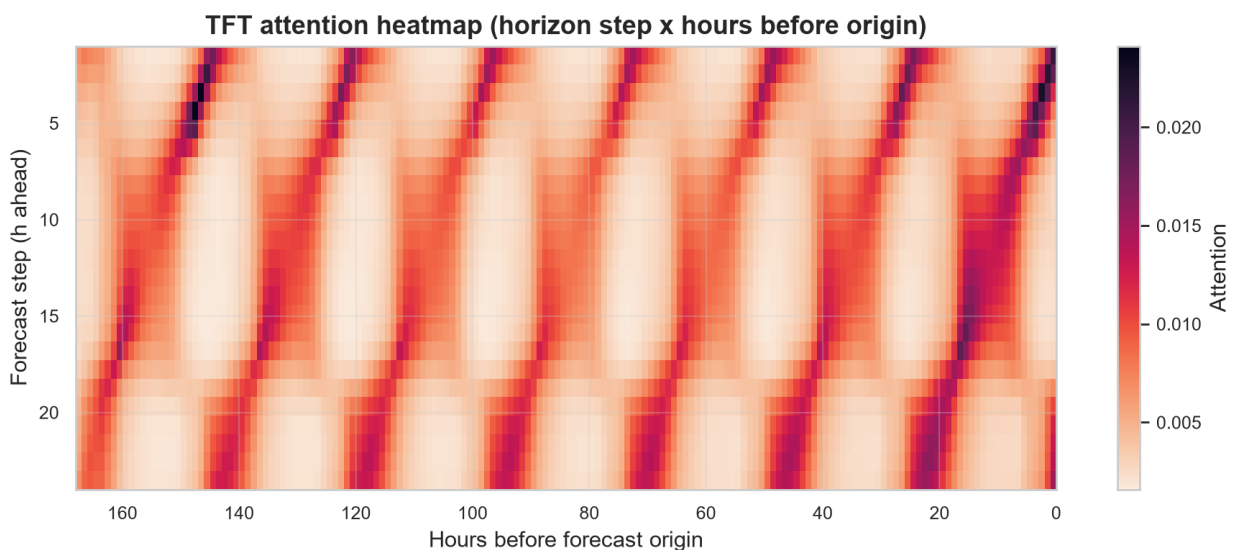


Figure 7. Attention by horizon step and past position.

6.2 Variable selection

The TFT variable-selection networks rank inputs by mean selection weight. Top encoder (past) inputs: `y_scaled` (0.69), `hour_cos` (0.05), `month_cos` (0.03), `hour_sin` (0.03), `is_weekend` (0.02). Top decoder (future) inputs: `hour_cos` (0.17), `is_weekend` (0.14), `month_cos` (0.13), `hour_sin` (0.10), `is_holiday` (0.09). Top static inputs: `use_Education` (0.14), `yearbuilt_missing` (0.13), `log_sqm_z` (0.13), `use_Office` (0.09), `yearbuilt_z` (0.08).

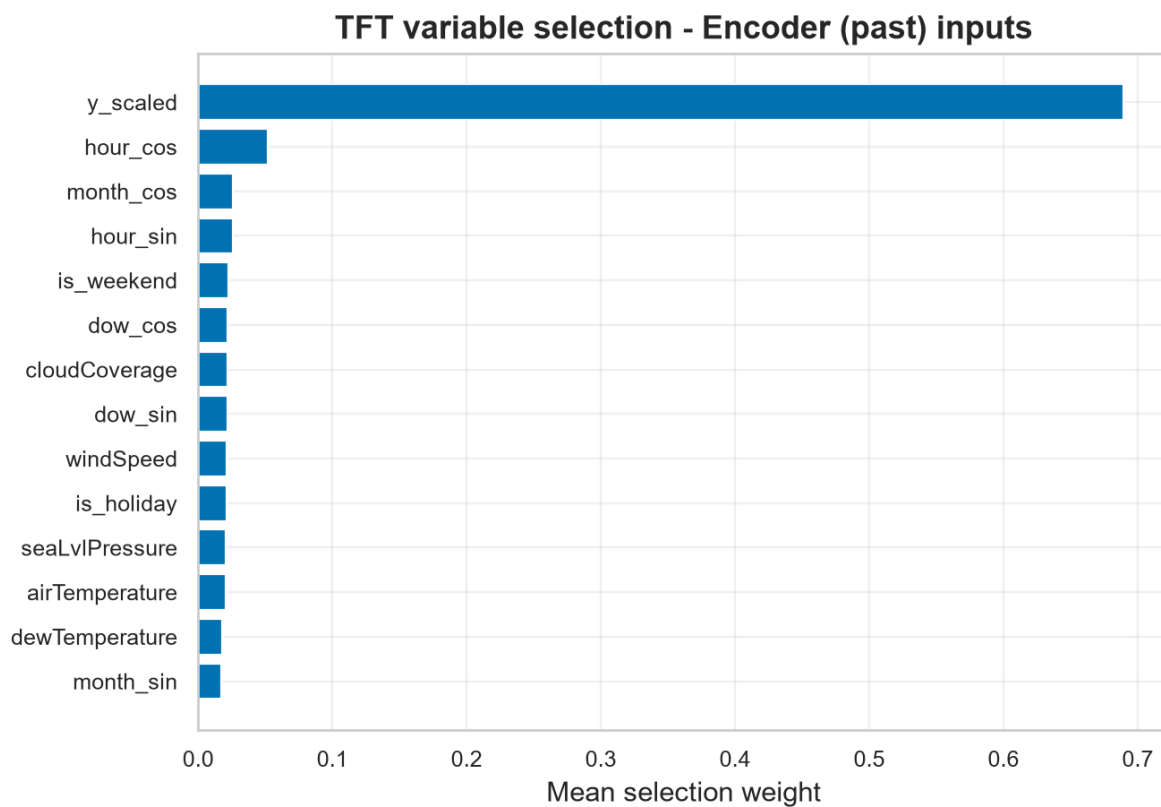


Figure 8. TFT encoder variable-selection importances.

6.3 Cross-check with gradient boosting

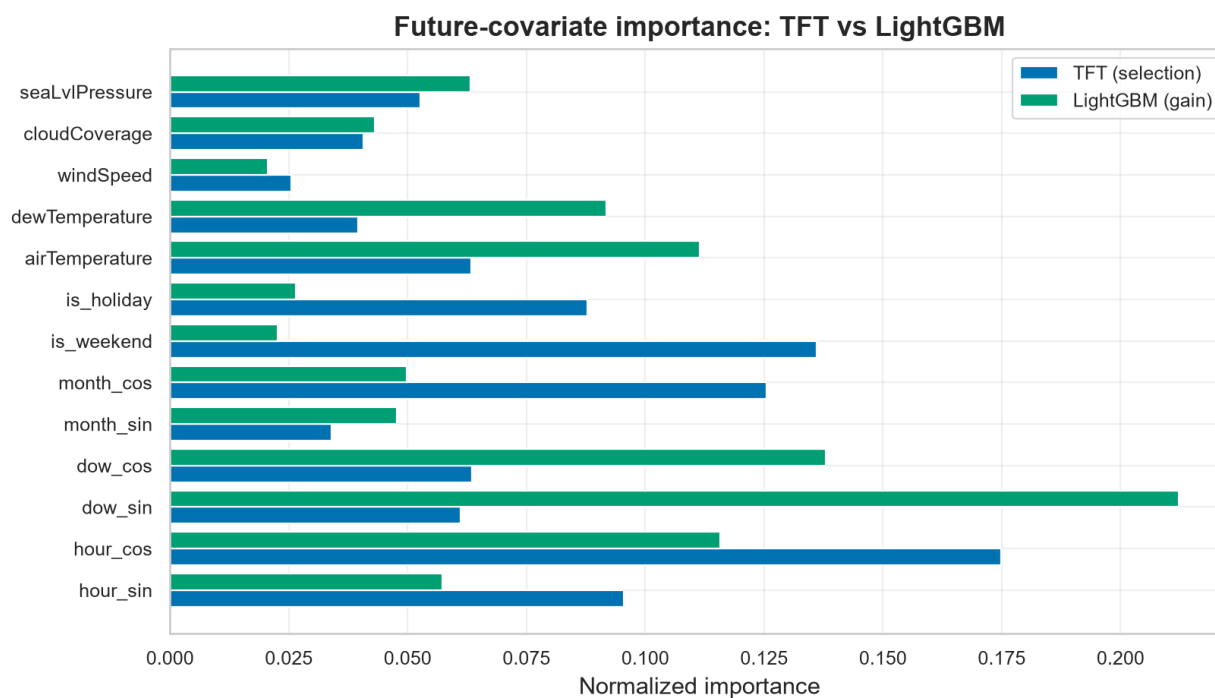


Figure 9. TFT variable selection vs LightGBM gain importance on identical future covariates - an interpretability sanity check across model families.

7. External Validation on ASHRAE GEPIII

model	MAE	RMSE	CV_RMSE	sMAPE	R2	MBE
seasonal_naive	28.129	148.209	56.916	16.473	0.955	0.018
transformer	44.918	161.899	62.173	26.178	0.947	-3.555
attn_lstm	41.643	172.688	66.316	24.291	0.939	19.776
tft	55.452	217.265	83.435	26.405	0.904	7.856
persistence	56.935	229.117	87.986	27.105	0.893	35.123
gru	73.51	294.256	113.001	30.812	0.824	48.664
lstm	71.458	317.095	121.772	36.326	0.796	26.757
cnn_lstm	104.161	421.958	162.042	37.6	0.639	41.689
tcn	120.205	529.748	203.435	35.771	0.43	76.761

Table 4. Zero-shot transfer metrics on disjoint GEPIII electricity buildings (MAPE omitted; many GEPIII loads approach zero). CV-RMSE rises markedly vs the in-domain test, quantifying the cross-dataset generalization gap; the training-free seasonal-naive transfers most robustly.

8. Discussion

The controlled setup lets us attribute differences to the model class. Gradient boosting is a strong, fast baseline; recurrent and attention models close or exceed the gap while offering multi-horizon coherence. Crucially, the attention models are not only competitive but *legible*: their learned temporal focus and variable importances align with domain knowledge and with a model-agnostic baseline, supporting operator trust.

9. Limitations

Experiments use a CPU-tractable subset (32 buildings) and the electricity meter; weather is treated as known over the horizon (a perfect-forecast assumption); and GEPIII shares provenance with BDG2, mitigated but not eliminated by overlap exclusion. Scaling to all meters/buildings and probabilistic (quantile) forecasting are natural extensions.

10. Conclusion

On a diverse BDG2 subset, attention-based deep learning delivers accurate day-ahead electricity forecasts while remaining interpretable. LightGBM leads on aggregate accuracy, and the TFT's attention and variable selection recover the daily/weekly periodicity and physically meaningful drivers of building load, narrowing the trust gap between accuracy and explainability.

References

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Appendix A. Reproducibility

All steps are driven by `config/config.yaml` (seed, subset, windowing, hyperparameters). Pipeline: `download -> preprocess -> features -> windowing -> eda -> train --all -> evaluate -> interpret -> external\_validation -> paper`.

A.1 Model size and training time

model	params	train_time_s
persistence	0	0
seasonal_naive	0	0
linear	0	6.1
lightgbm	0	27.5
lstm	222593	563.9
gru	169473	1428
tcn	73793	1128.9
cnn_lstm	142593	665.3
transformer	103233	5296.1
attn_lstm	263681	669.2
tft	297441	9430.5